Practical no. 2

Aim: Write the program with different features of C#
a. Function Overloading
b. Inheritance (all types)
c. Constructor overloading
d. Interfaces
e. Using Delegates and events
f. Exception handling

Software Required:

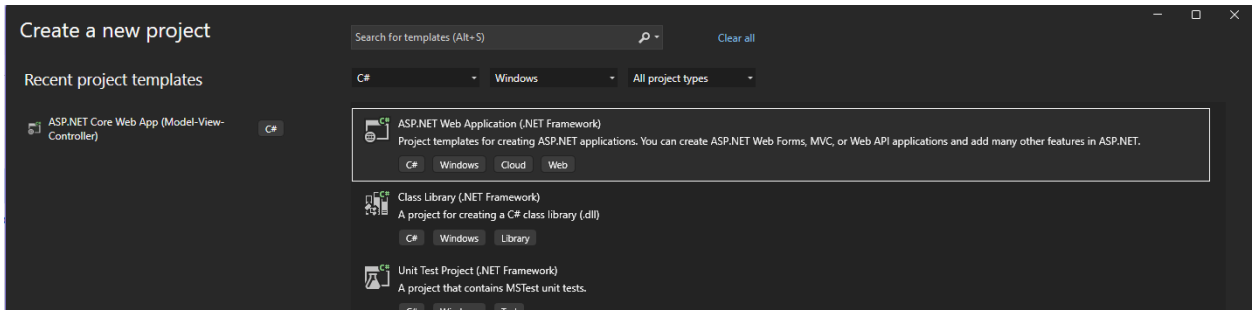
Visual Studio 2026

Procedure:

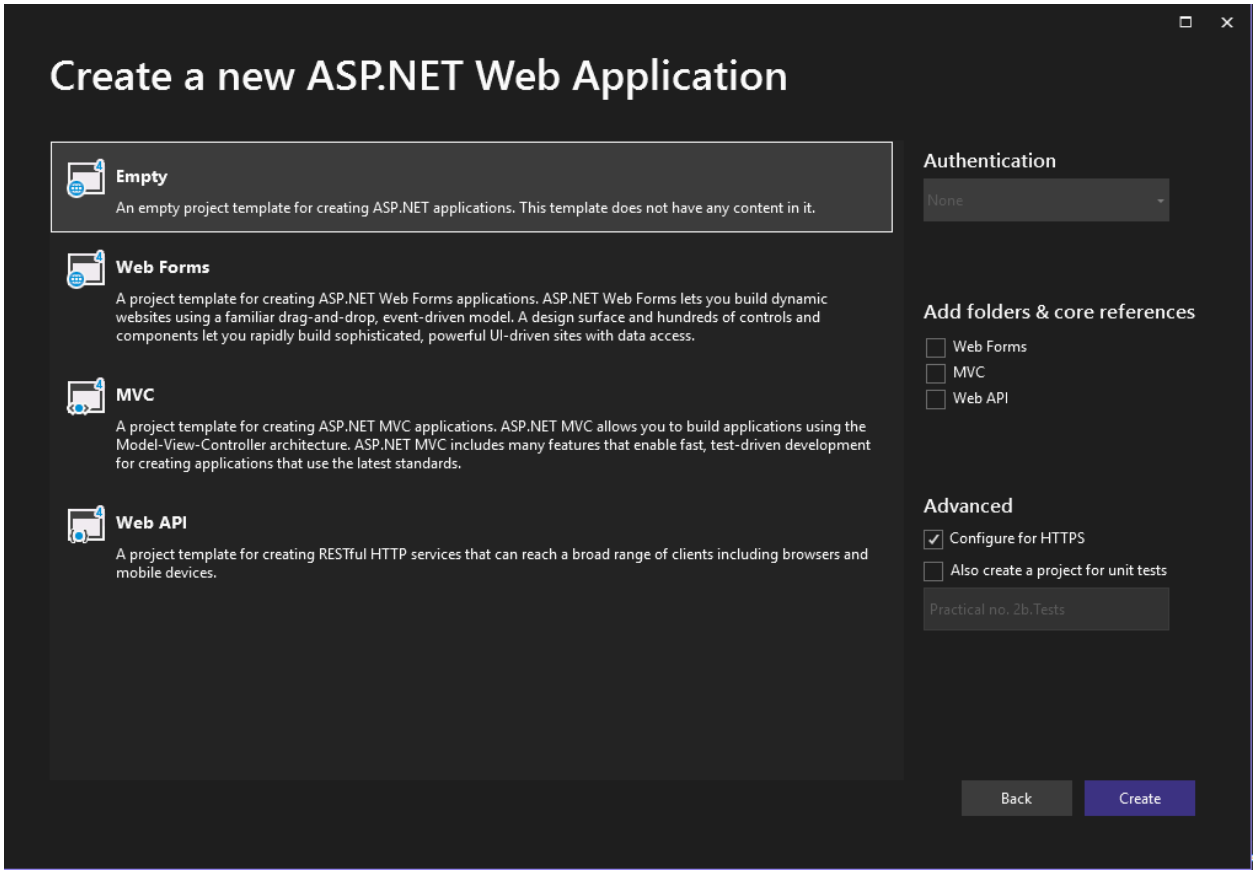
a. Function Overloading

Step 1: Create New Project

1. Open Visual Studio.
2. Click **Create a new project**.
3. Select **ASP.NET Web Application (.NET Framework)**.



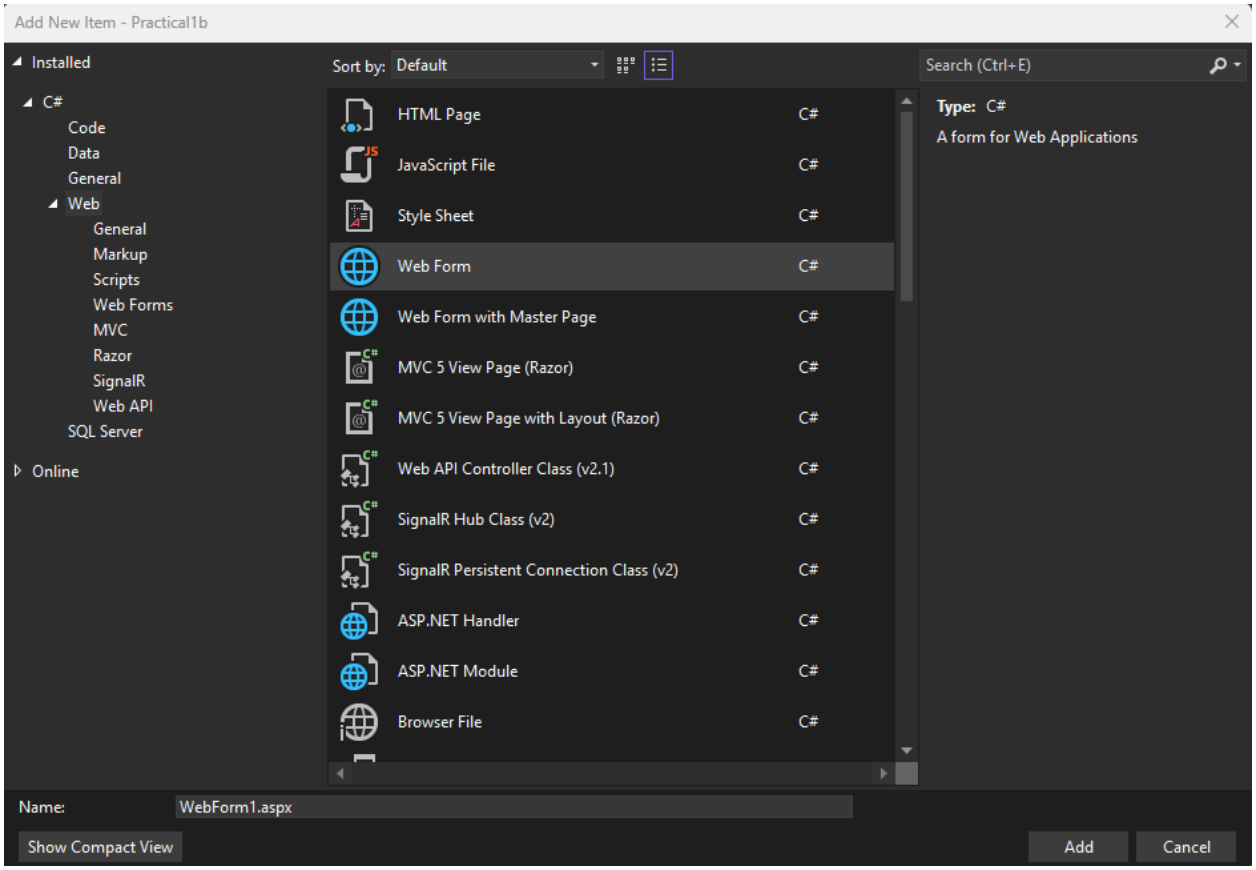
4. Click **Next**.
5. Enter Project Name: **Practical2**
6. Click **Create**.
7. Select **Empty**.



8. Click **Create**.

Step 2: Add a Web Form

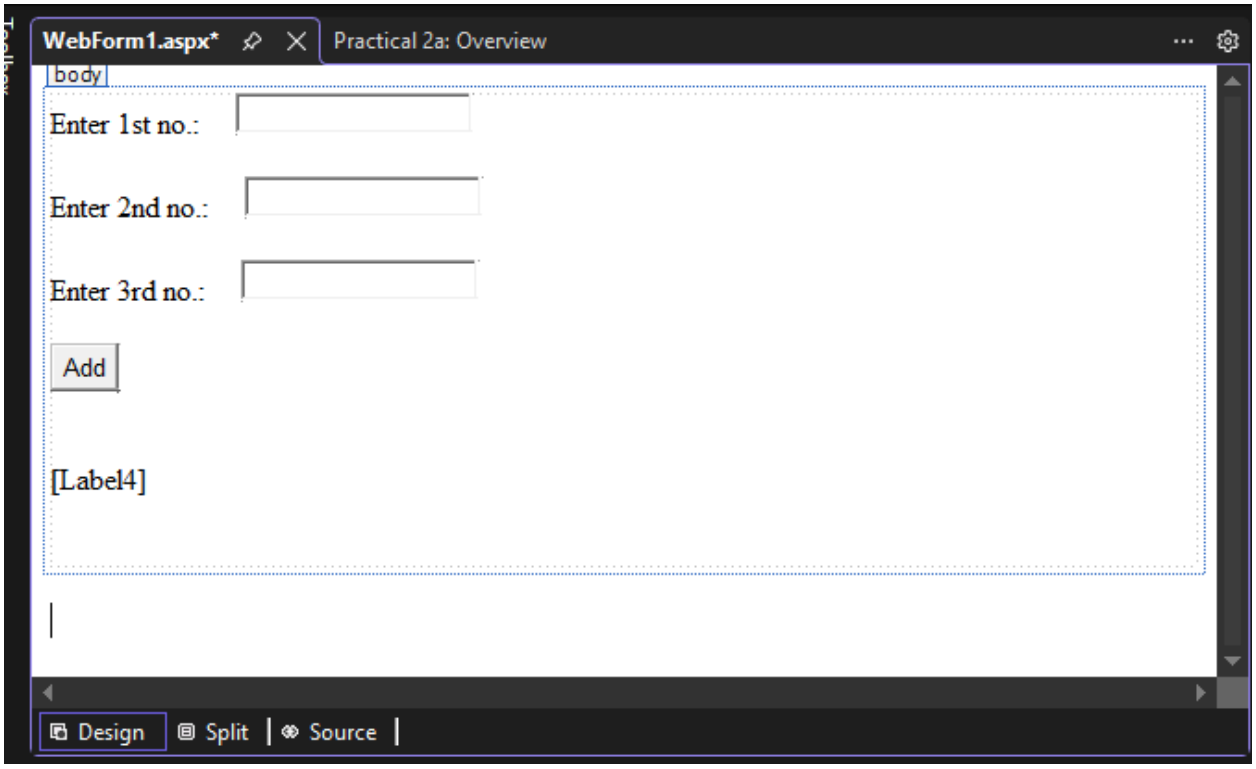
1. In Solution Explorer, right-click the project **Practical2**
2. Select Add → New Item.
3. Select Web Form.
4. Click **Add**.



Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. TextBox
2. Button
3. Label



Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_2a
{
    public partial class WebForm1 : System.Web.UI.Page
    {
        public int Add(int a, int b)
        {
            return a + b;
        }

        public int Add(int a, int b, int c)
        {
```

```
        return a + b + c;
    }
    protected void Page_Load(object sender, EventArgs e)
    {

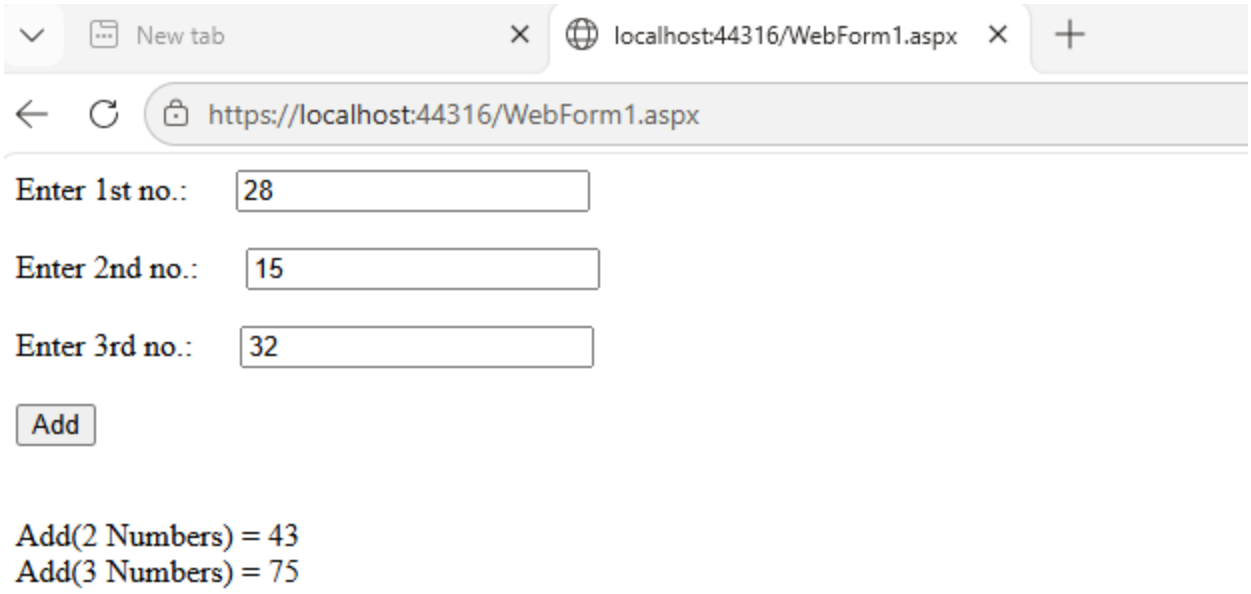
    }

    protected void Button1_Click(object sender, EventArgs e)
    {
        int a = Convert.ToInt32(TextBox1.Text);
        int b = Convert.ToInt32(TextBox2.Text);
        int c = Convert.ToInt32(TextBox3.Text);

        Label4.Text =
        "Add(2 Numbers) = " + Add(a, b) +
        "<br/>Add(3 Numbers) = " + Add(a, b, c);
    }
  }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press F5.
4. Browser window will open.
5. Enter three numbers in TextBox .
6. Click **Add**.



b. Inheritance (all types)

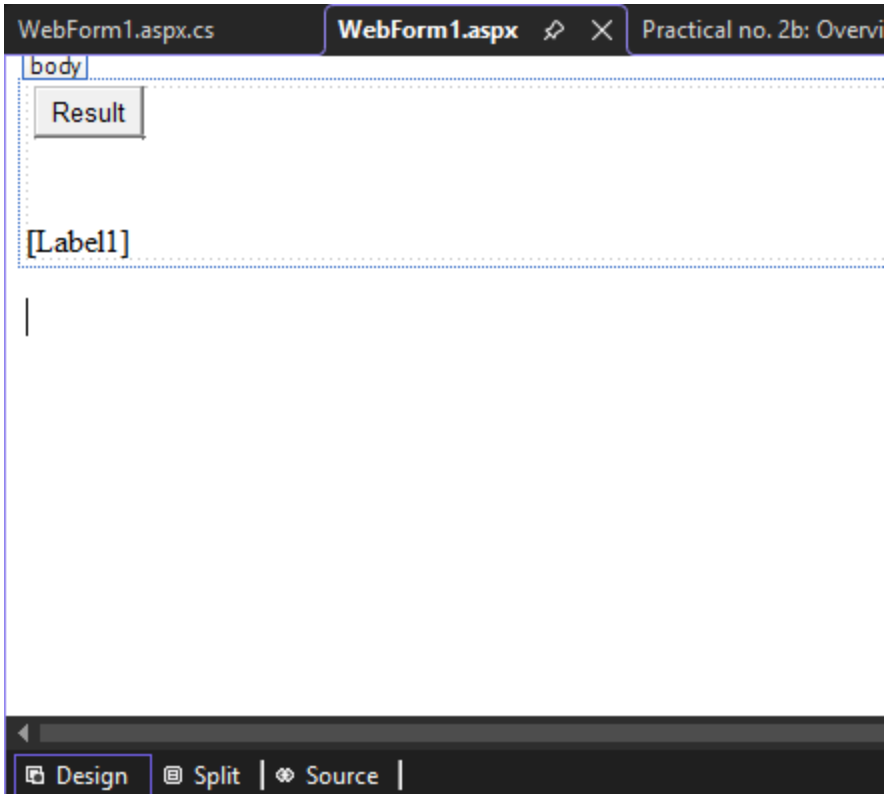
Procedure:

Step 1 and Step 2 are the same as the previous practical.

Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. Button
2. Label



Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_no._2b
{
   public partial class WebForm1 : System.Web.UI.Page
   {

      // SINGLE INHERITANCE

      class Person
      {
```

```
    public string ShowName()
    {
        return "Person Class";
    }
}

class Student : Person
{
    public string Study()
    {
        return "Student is Studying";
    }
}
```

// MULTILEVEL INHERITANCE

```
class GrandFather
{
    public string Grand()
    {
        return "GrandFather";
    }
}

class Father : GrandFather
{
    public string Dad()
    {
        return "Father";
    }
}

class Son : Father
{
    public string Child()
    {
        return "Son";
    }
```

```
}
```

// HIERARCHICAL INHERITANCE

```
class Animal
{
   public string Eat()
   {
      return "Animal Eats Food";
   }
}

class Dog : Animal
{
   public string Bark()
   {
      return "Dog Barks";
   }
}

class Cat : Animal
{
   public string Meow()
   {
      return "Cat Meows";
   }
}
```

//MULTIPLE INHERITANCE

```
interface ISports
{
   string Play();
}

class Teacher
{
   public string Teach()
```

```
    {
        return "Teacher Teaches";
    }
}

class SportsTeacher : Teacher, ISports
{
    public string Play()
    {
        return "Sports Teacher Plays Cricket";
    }
}



protected void Page_Load(object sender, EventArgs e)
{

}

protected void Button1_Click(object sender, EventArgs e)
{
    Label1.Text = "";

    // Single Inheritance

    Student s = new Student();

    Label1.Text += "<b>SINGLE INHERITANCE</b><br/>";
    Label1.Text += s.ShowName() + "<br/>";
    Label1.Text += s.Study() + "<br/><br/>";

    // Multilevel Inheritance

    Son son = new Son();

    Label1.Text += "<b>MULTILEVEL INHERITANCE</b><br/>";
    Label1.Text += son.Grand() + "<br/>";
    Label1.Text += son.Dad() + "<br/>";
    Label1.Text += son.Child() + "<br/><br/>";
```

```
        // Hierarchical Inheritance

        Dog d = new Dog();
        Cat c = new Cat();

        Label1.Text += "<b>HIERARCHICAL INHERITANCE</b><br/>";
        Label1.Text += d.Eat() + "<br/>";
        Label1.Text += d.Bark() + "<br/>";
        Label1.Text += c.Eat() + "<br/>";
        Label1.Text += c.Meow() + "<br/><br/>";

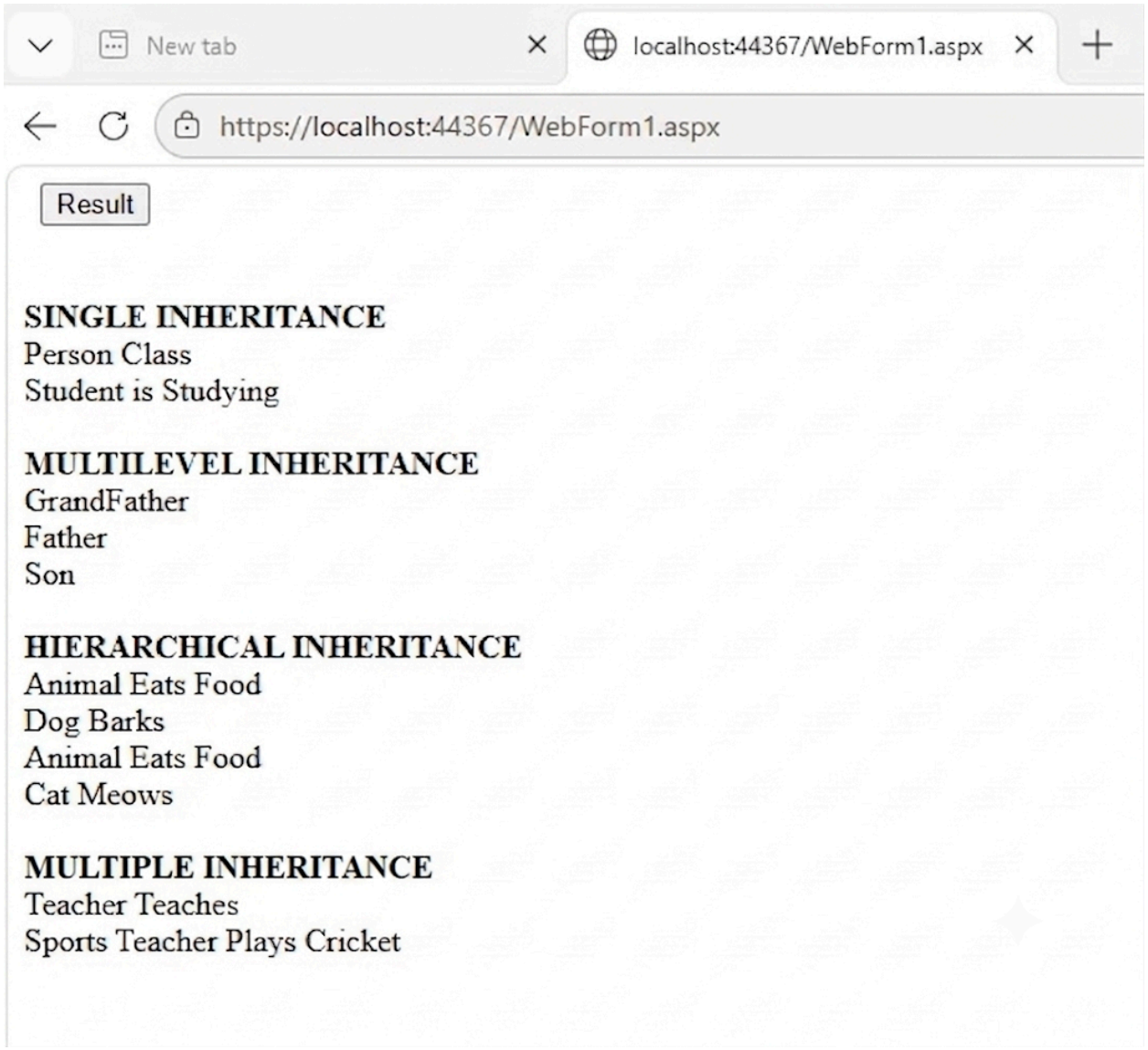
        // Multiple Inheritance

        SportsTeacher st = new SportsTeacher();

        Label1.Text += "<b>MULTIPLE INHERITANCE</b><br/>";
        Label1.Text += st.Teach() + "<br/>";
        Label1.Text += st.Play();
    }
  }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press **F5**.
4. Browser window will open.
5. Enter three numbers in TextBox .
6. Click **Result**.



c. Constructor overloading

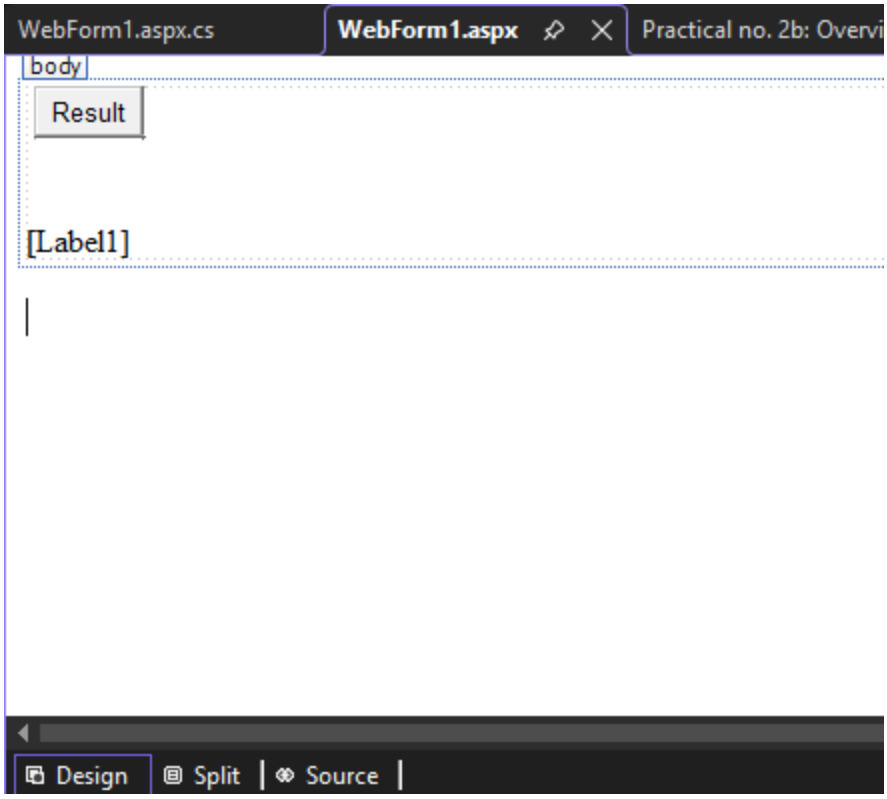
Procedure:

Step 1 and Step 2 are the same as the previous practical.

Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. Button
2. Label



Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_no._2b
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      class Student
      {
         public string msg;

         public Student()
         {
            msg = "Default Constructor";
```

```
        }

        public Student(string name)
        {
            msg = "Name = " + name;
        }

        public Student(string name, int age)
        {
            msg = "Name = " + name +
                  ", Age = " + age;
        }
    }

    protected void Page_Load(object sender, EventArgs e)
    {

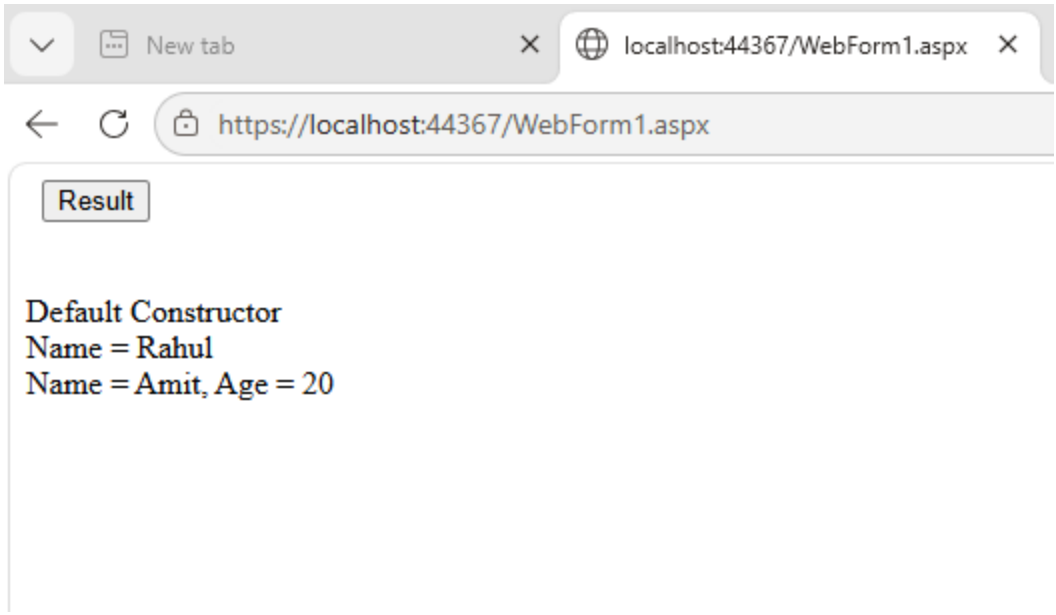
    }

    protected void Button1_Click(object sender, EventArgs e)
    {
        Student s1 = new Student();
        Student s2 = new Student("Rahul");
        Student s3 = new Student("Amit", 20);

        Label1.Text =
            s1.msg + "<br/>" +
            s2.msg + "<br/>" +
            s3.msg;
    }
  }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press **F5**.
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5. Enter three numbers in TextBox .
6. Click **Result**.



d. Interfaces

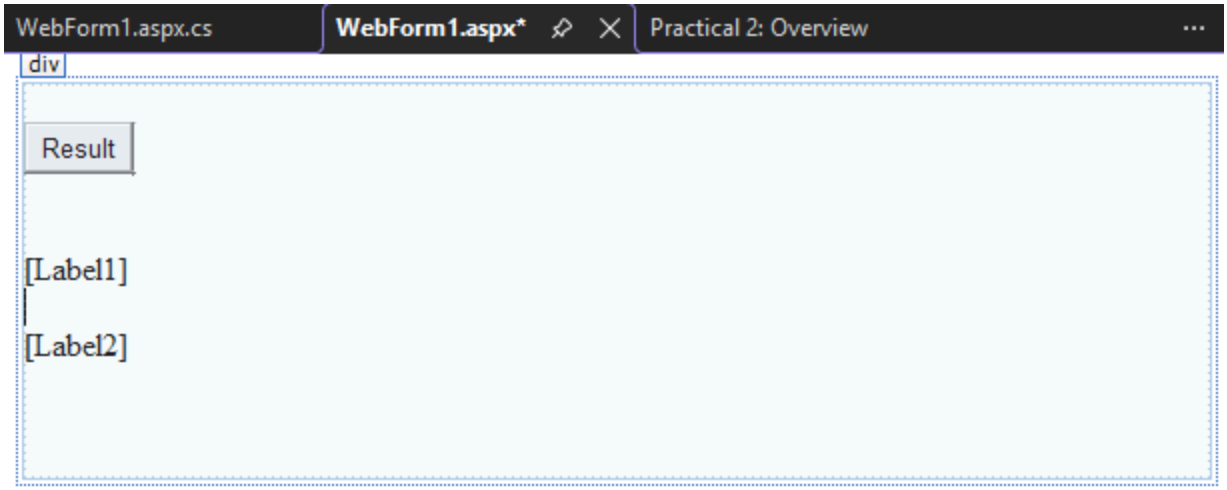
Procedure:

Step 1 and Step 2 are the same as the previous practical.

Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. Button
2. Label





Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Drawing;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_2
{
   public partial class WebForm1 : System.Web.UI.Page
  {
  public class shape
  {
      public int x, y, z;
      public void getSide(int length, int breadth, int width)
```

```
        {
            x = length;
            y = breadth;
            z = width;
        }
    }
    public interface IPolygon
    {
        int CalculateArea();
    }
    public interface IVolume
    {
        int getVolume();
    }
    class Rectangle : shape, IPolygon, IVolume
    {
        public int CalculateArea()
        {
            return (x * y);
        }
        public int getVolume()
        {
            return (x * y * z);
        }
    }

        protected void Page_Load(object sender, EventArgs e)
        {

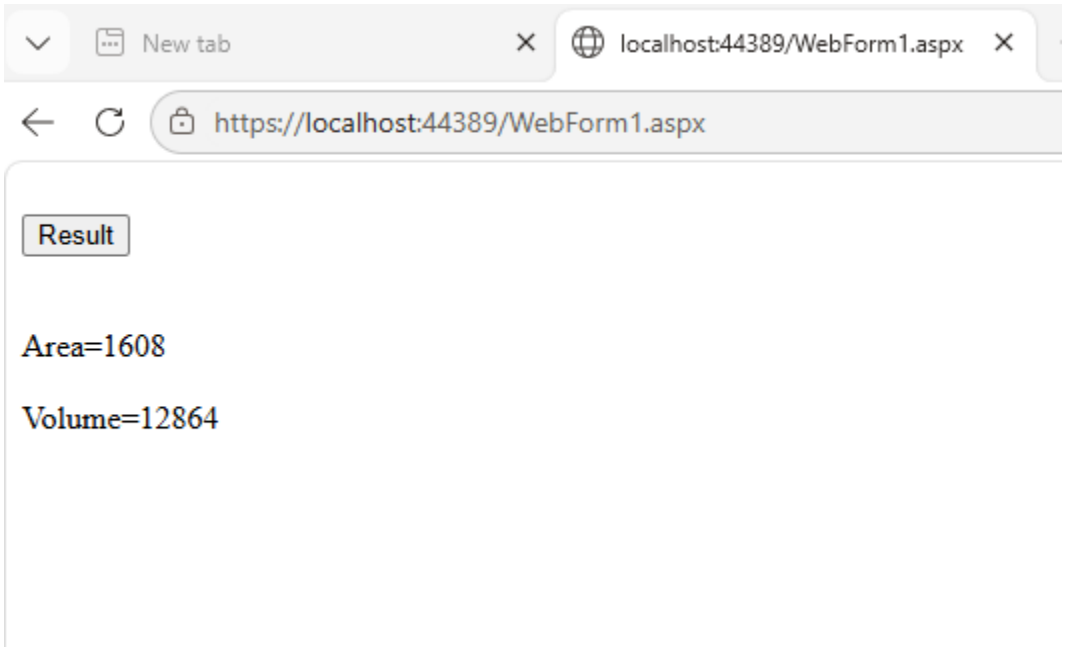
        }

        protected void Button1_Click(object sender, EventArgs e)
        {
            Rectangle r = new Rectangle();
            r.getSide(24, 67, 8);
            Label1.Text = "Area=" + r.CalculateArea() +"<br/>" +"Volume="+ r.getVolume();

        }
    }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press **F5**.
4. Browser window will open.
5. Enter three numbers in TextBox .
6. Click **Result**.



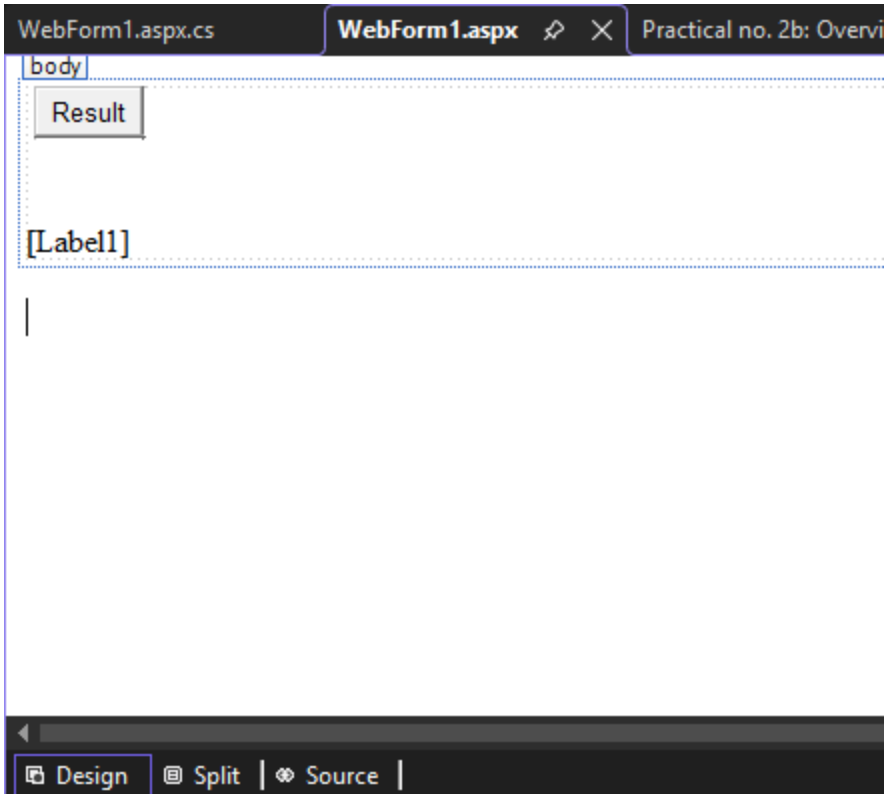
e. Using Delegates and events
Procedure:

Step 1 and Step 2 are the same as the previous practical.

Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. Button
2. Label



Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_2
{
   public partial class WebForm1 : System.Web.UI.Page
   {
      // Delegate Declaration
      public delegate void ResultHandler();

      // Event Declaration
      public event ResultHandler ResultDeclared;
```

```
    protected void Page_Load(object sender, EventArgs e)
    {

    }

    protected void Button1_Click(object sender, EventArgs e)
    {
        // Subscribe Event
        ResultDeclared += ShowMessage;

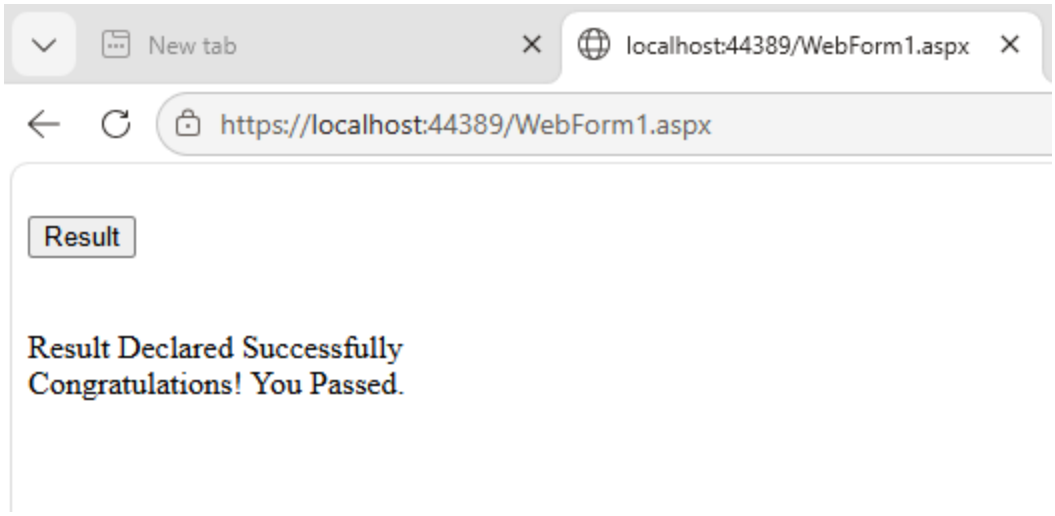
        Label1.Text = "Result Declared Successfully<br/>";

        // Raise Event
        ResultDeclared();
    }

    public void ShowMessage()
    {
        Label1.Text += "Congratulations! You Passed.";
    }
  }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press **F5**.
4. Browser window will open.
5. Enter three numbers in TextBox .
6. Click **Result.**



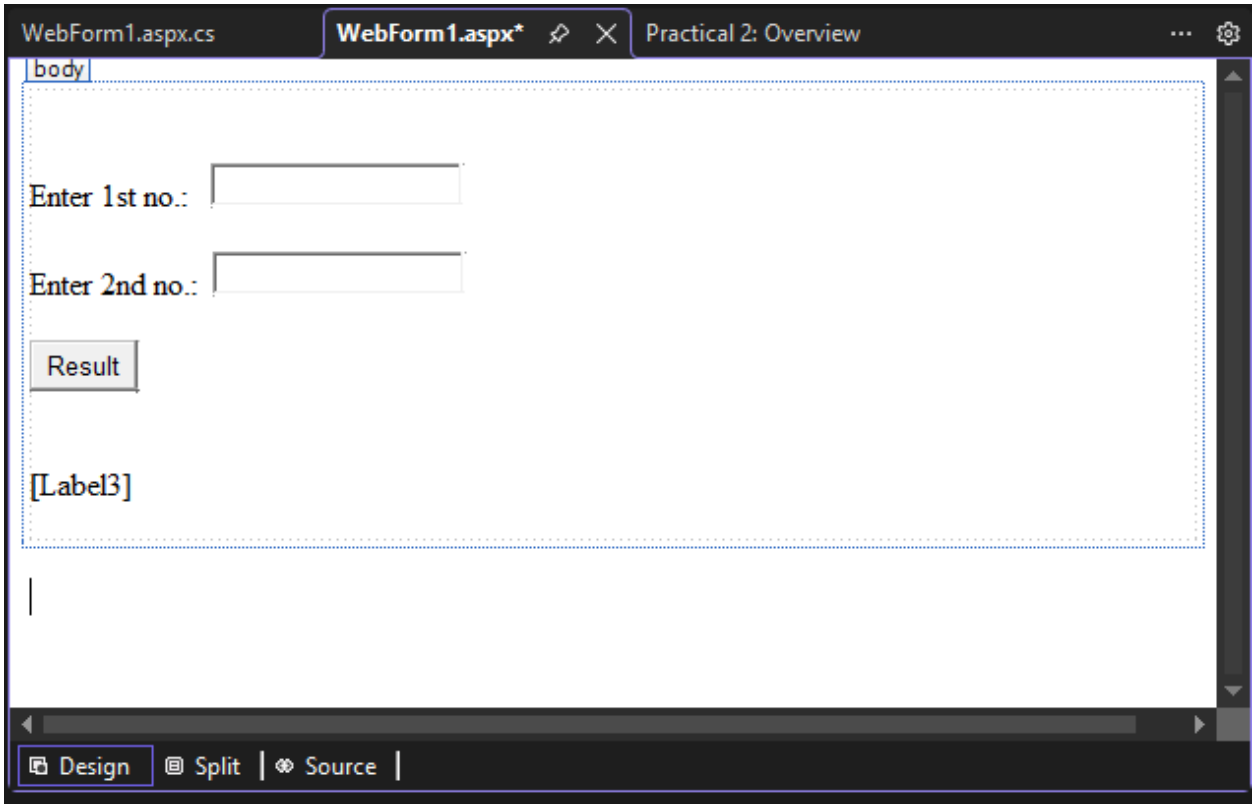
f. Exception Handling
Procedure:

Step 1 and Step 2 are the same as the previous practical.

Step 3: Design Web Form

Open **WebForm1.aspx** and add the following controls:

1. TextBox
2. Button
3. Label



Step 4: Write C# Code

Open **WebForm1.aspx.cs** and write the following code.

WebForm1.aspx.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace Practical_2
{
   public partial class WebForm1 : System.Web.UI.Page
   {

      protected void Page_Load(object sender, EventArgs e)
      {

      }
```

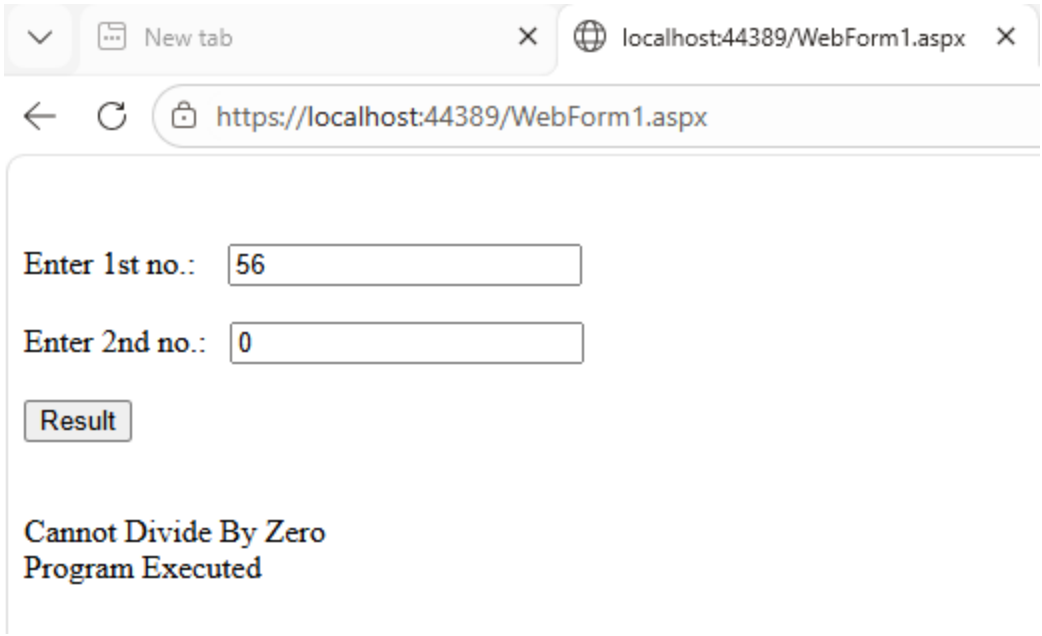
```
    protected void Button1_Click(object sender, EventArgs e)
    {
        try
        {
            int a = Convert.ToInt32(TextBox1.Text);
            int b = Convert.ToInt32(TextBox2.Text);

            int result = a / b;

            Label3.Text = "Result = " + result;
        }
        catch (DivideByZeroException)
        {
            Label3.Text =
            "Cannot Divide By Zero";
        }
        catch (FormatException)
        {
            Label3.Text =
            "Invalid Input";
        }
        finally
        {
            Label3.Text +=
            "<br/>Program Executed";
        }
    }
  }
}
```

Step 5: Run the Application

1. Save all files.
2. Click Build →Build Solution.
3. Press **F5**.
4. Browser window will open.
5. Enter three numbers in TextBox .
6. Click **Result**


New tab
localhost:44389/WebForm1.aspx
https://localhost:44389/WebForm1.aspx
Enter 1st no.:
56
Enter 2nd no.:
0
Result
Cannot Divide By Zero
Program Executed